Routing And Switching Project Via Cisco Packet Tracer

NAME: Huzaifa Habib

Email: huzaifa.habib842@gmail.com

Linkedin: https://www.linkedin.com/in/huzaifa-h-9a2725301

Github: https://github.com/HuzaifaHabib1

Contents

1. Executive Summary 2
2. Problem Statement 2
3. Network Design and Hardware Setup 3
4. Implementation Strategy 3
4.1 Dynamic Routing (OSPF) 3
4.2 Static Route Override 3
5. Verification and Connectivity Report 9
Test 1: Full Connectivity (PC0 to PC2) 9
Test 2: Inter-Branch Connectivity (PC1 to PC2) 10
Test 3: Path Verification (Traceroute) 10
6. Challenges and Solutions 12
1. Interface Availability 12
2. Clock Rates 12
7. Conclusion 12

1. Executive Summary

This project demonstrates the design and simulation of a WAN topology for an organization consisting of a Head Office and four branch offices. The primary objective was to establish full connectivity using Open Shortest Path First (OSPF) as the dynamic routing protocol. Additionally, a specific traffic engineering requirement was implemented to force traffic from the Head Office to Branch 2 via an indirect path using a Static Route. The simulation was successfully carried out in Cisco Packet Tracer, confirming both the mesh connectivity and the required path manipulation.

2. Problem Statement

The scenario required connecting a Head Office to four distinct branches in a partial mesh topology. While OSPF was required for general connectivity and redundancy, a specific constraint was placed on the traffic flow: Data sent from the Head Office to Branch 2 was required to bypass the direct link and instead travel via Branch 1 (Router 1 → Router 0 → Router 2). This required overriding the default OSPF "shortest path" behavior.

3. Network Design and Hardware Setup

The network topology consists of five routers and their respective LANs.

Hardware Configuration: During the initial setup, a hardware limitation was identified. The standard routers did not have enough built-in serial ports to support the required mesh connections for the Head Office and Branch 2 routers.

- **Action Taken:** I physically upgraded Router 1 (Head Office) and Router 2 (Branch 2) by powering them down and installing the **NM-1SS** expansion module. This provided the necessary Serial6/0 interface to complete the topology.

IP Addressing Scheme:

- **WAN Links:** 192.168.1.0/24 through 192.168.6.0/24.
- **LAN Networks:** 192.168.7.0/24 (Branch 1), 192.168.8.0/24 (Branch 2), and 192.168.9.0/24 (Branch 4).

4. Implementation Strategy

4.1 Dynamic Routing (OSPF)

I configured OSPF Process ID 1 on all routers and advertised all directly connected networks into Area 0 (Backbone). This established the baseline connectivity, allowing all branches to communicate with each other and the Head Office.

4.2 Static Route Override

To meet the specific requirement for the Head Office to Branch 2 traffic, I utilized the concept of Administrative Distance (AD).

- OSPF has a default AD of 110.
- Static Routes have a default AD of 1.
- **Configuration:** I entered the command ip route 192.168.8.0 255.255.255.0 192.168.1.2 on the Head Office router. Because the AD of the static route is lower, the router installs this path into the routing table, effectively ignoring the direct OSPF link for this specific destination.

Branch 1 Router Configuration:

CopyRouter0

Physical | Config | CLI

IOS Comm

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface se2/0
Router(config-if)#ip address 192.168.1.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#interface s
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
e3/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.7.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#
00:08:58: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.5.1 on Serial2/0 from LOADING to FULL, Loading Done
network 192.168.7.0 0.0.0.255 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#exit
Router>sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, Serial2/0
C    192.168.2.0/24 is directly connected, Serial3/0
O    192.168.3.0/24 [110/128] via 192.168.1.1, 00:07:42, Serial2/0
                    [110/128] via 192.168.2.2, 00:07:42, Serial3/0
O    192.168.4.0/24 [110/128] via 192.168.2.2, 00:06:21, Serial3/0
O    192.168.5.0/24 [110/128] via 192.168.1.1, 00:04:31, Serial2/0
O    192.168.6.0/24 [110/192] via 192.168.1.1, 00:02:31, Serial2/0
                    [110/192] via 192.168.2.2, 00:02:31, Serial3/0
C    192.168.7.0/24 is directly connected, FastEthernet0/0
O    192.168.8.0/24 [110/65] via 192.168.2.2, 00:07:52, Serial3/0
O    192.168.9.0/24 [110/129] via 192.168.1.1, 00:02:31, Serial2/0
Router>
```

Head Office Router Configuration:

CopyRouter1

Physical | Config | CLI

IO

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface se2/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#interface se6/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial6/0, changed state to down
Router(config-if)#exit
Router(config)#interface se3/0
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#router ospfl
                         ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#ip route 192.168.8.0 255.255.255.0 192.168.1.2
Router(config)#exit
Router#
```

```
Router>sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, Serial2/0
O    192.168.2.0/24 [110/128] via 192.168.1.2, 00:08:40, Serial2/0
                    [110/128] via 192.168.3.2, 00:08:40, Serial6/0
C    192.168.3.0/24 is directly connected, Serial6/0
O    192.168.4.0/24 [110/128] via 192.168.3.2, 00:07:28, Serial6/0
C    192.168.5.0/24 is directly connected, Serial3/0
O    192.168.6.0/24 [110/128] via 192.168.5.2, 00:03:37, Serial3/0
O    192.168.7.0/24 [110/65] via 192.168.1.2, 00:12:04, Serial2/0
S    192.168.8.0/24 [1/0] via 192.168.1.2
O    192.168.9.0/24 [110/65] via 192.168.5.2, 00:03:48, Serial3/0
Router>
```

Branch 2 Router Configuration:

CopyRouter2

Physical | Config | CLI

IOS C

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.8.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface se2/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#interface se
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
6/0
Router(config-if)#ip address 192.168.3.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial6/0, changed state to up
exit
Router(config)#interface se3/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial6/0, changed state to up
ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 192.168.
00:12:05: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.7.1 on Serial2/0 from LOADING to FULL, Loading Done
8.0 0.0.0.255 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 192.168.4.0
00:12:34: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.5.1 on Serial6/0 from LOADING to FULL, Loading Done
 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
```

```
sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.1.0/24 [110/128] via 192.168.2.1, 00:09:59, Serial2/0
                    [110/128] via 192.168.3.1, 00:09:59, Serial6/0
C    192.168.2.0/24 is directly connected, Serial2/0
C    192.168.3.0/24 is directly connected, Serial6/0
C    192.168.4.0/24 is directly connected, Serial3/0
O    192.168.5.0/24 [110/128] via 192.168.3.1, 00:06:57, Serial6/0
O    192.168.6.0/24 [110/128] via 192.168.4.2, 00:06:28, Serial3/0
O    192.168.7.0/24 [110/65] via 192.168.2.1, 00:10:29, Serial2/0
C    192.168.8.0/24 is directly connected, FastEthernet0/0
O    192.168.9.0/24 [110/129] via 192.168.3.1, 00:04:47, Serial6/0
                    [110/129] via 192.168.4.2, 00:04:47, Serial3/0
Router#
```

Branch 3 Router Configuration:

CopyRouter3

Physical | Config | CLI

IOS

```
Router(config)#interface se3/0
Router(config-if)#ip address 192.168.4.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
exit
Router(config)#interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
 se2/0
Router(config-if)#ip address 192.168.6.1 255.255.255.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.4.0 0.0.0.255 area 0
Router(config-router)#network 192.168.6.0
00:14:47: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.8.1 on Serial3/0 from LOADING to FULL, Loading Done
0.0.0.255 area 0
Router(config-router)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

00:17:46: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.9.1 on Serial2/0 from LOADING to FULL, Loading Done

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.1.0/24 [110/192] via 192.168.4.1, 00:05:36, Serial3/0
                    [110/192] via 192.168.6.2, 00:05:36, Serial2/0
O    192.168.2.0/24 [110/128] via 192.168.4.1, 00:08:31, Serial3/0
O    192.168.3.0/24 [110/128] via 192.168.4.1, 00:08:31, Serial3/0
C    192.168.4.0/24 is directly connected, Serial3/0
O    192.168.5.0/24 [110/128] via 192.168.6.2, 00:05:36, Serial2/0
C    192.168.6.0/24 is directly connected, Serial2/0
O    192.168.7.0/24 [110/129] via 192.168.4.1, 00:08:31, Serial3/0
O    192.168.8.0/24 [110/65] via 192.168.4.1, 00:08:31, Serial3/0
O    192.168.9.0/24 [110/65] via 192.168.6.2, 00:05:36, Serial2/0
```

Branch 4 Router Configuration:

CopyRouter4

Physical | Config | CLI

IOS

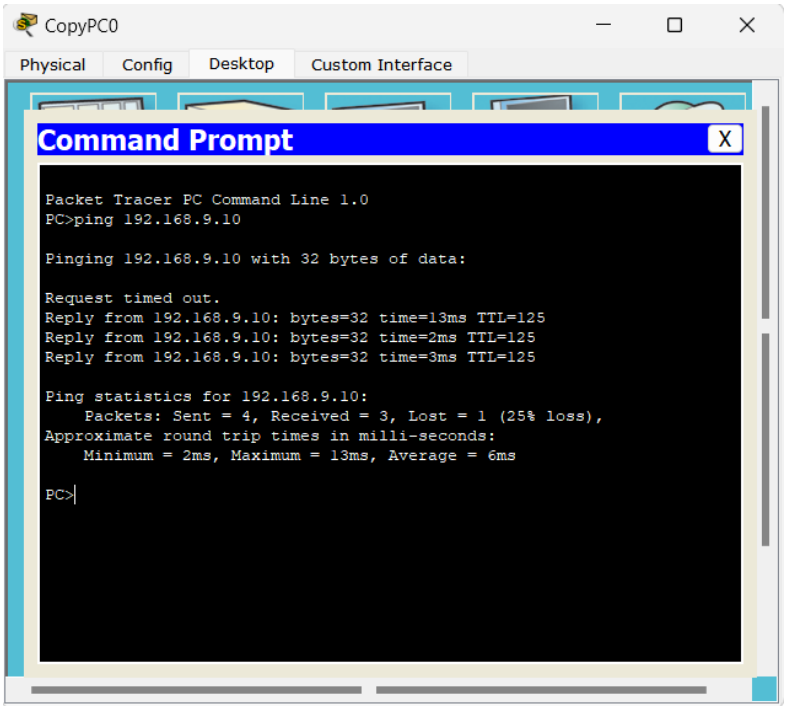
```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface se3/0
Router(config-if)#ip address 192.168.5.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
exit
Router(config)#interface se
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
2/0
Router(config-if)#ip address 192.168.6.2 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#interface fa
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
0/0
Router(config-if)#ip address 192.168.9.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.9.0 0.0.0.255 area 0
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#network
00:17:27: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.5.1 on Serial3/0 from LOADING to FULL, Loading Done
 192.168.6.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.1.0/24 [110/128] via 192.168.5.1, 00:00:35, Serial3/0
O    192.168.2.0/24 [110/192] via 192.168.5.1, 00:00:14, Serial3/0
                    [110/192] via 192.168.6.1, 00:00:14, Serial2/0
O    192.168.3.0/24 [110/128] via 192.168.5.1, 00:00:35, Serial3/0
O    192.168.4.0/24 [110/128] via 192.168.6.1, 00:00:14, Serial2/0
C    192.168.5.0/24 is directly connected, Serial3/0
C    192.168.6.0/24 is directly connected, Serial2/0
O    192.168.7.0/24 [110/129] via 192.168.5.1, 00:00:35, Serial3/0
O    192.168.8.0/24 [110/129] via 192.168.5.1, 00:00:14, Serial3/0
                    [110/129] via 192.168.6.1, 00:00:14, Serial2/0
C    192.168.9.0/24 is directly connected, FastEthernet0/0
Router#
```

5. Verification and Connectivity Report

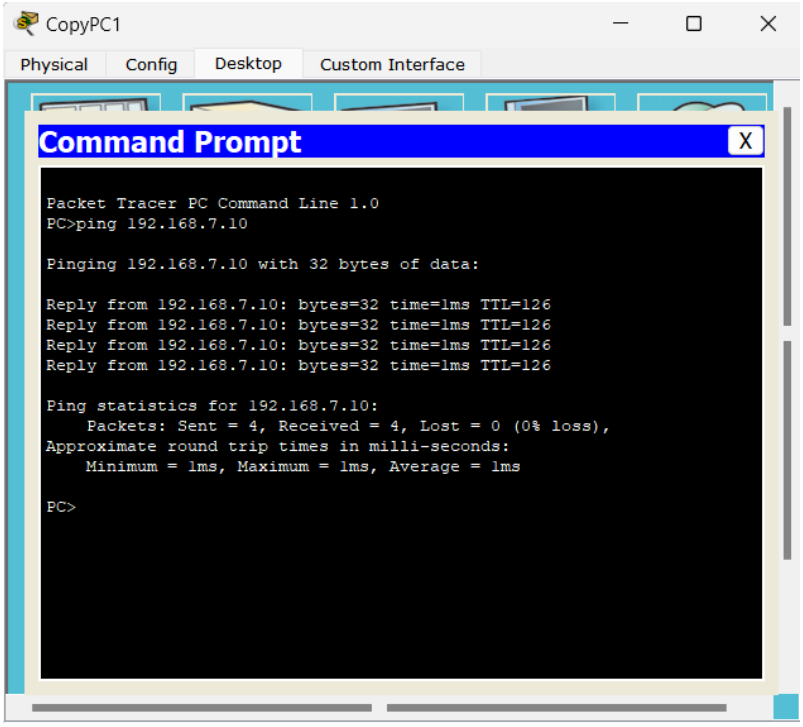
To validate the network, I performed ping tests between various branches and a traceroute to confirm the path manipulation.

Test 1: Full Connectivity (PC0 to PC2)

I pinged from Branch 1 (PC0) to Branch 4 (PC2) to verify that the OSPF backbone was functioning correctly across the network. **Status:** Successful.

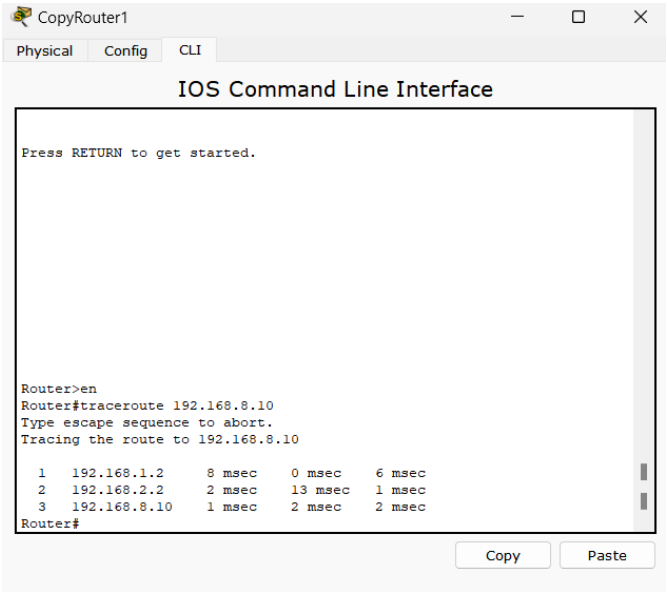


Test 2: Inter-Branch Connectivity (PC1 to PC2) I pinged from Branch 2 (PC1) to Branch 4 (PC2). This confirms that the mesh topology is routing traffic correctly between indirect branches. **Status:** Successful.

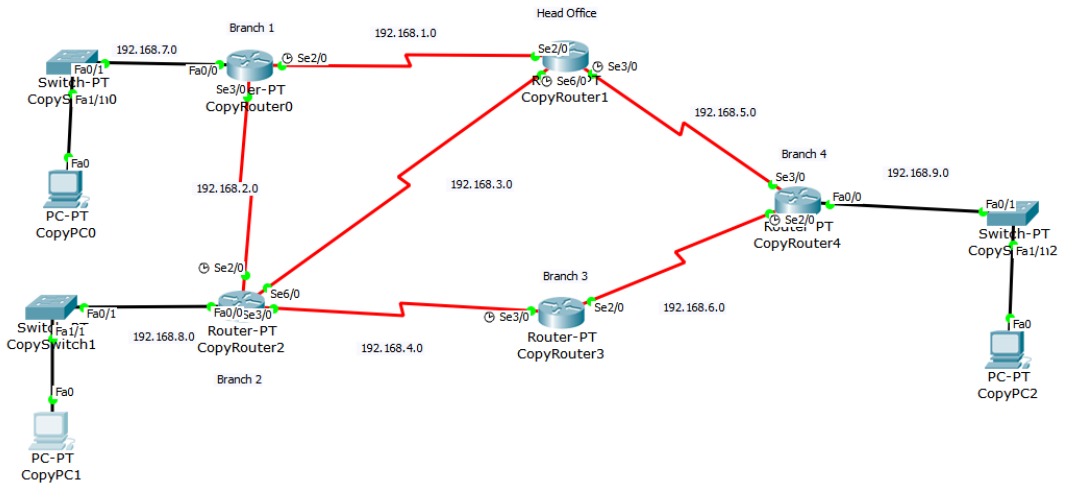


Test 3: Path Verification (Traceroute) This was the critical test to verify the complex problem requirement. I ran a traceroute from the Head Office Router to the Branch 2 PC (192.168.8.10).

- **Observation:** The trace showed the first hop as 192.168.1.2 (Branch 1) rather than the direct link.
- **Conclusion:** The static route is successfully overriding OSPF.



NETWORK DIAGRAM/TOPOLOGY



6. Challenges and Solutions

1. Interface Availability**:** As mentioned, the default routers lacked ports. Adding the NM-1SS modules required careful interface naming (e.g., using Serial6/0), which I had to account for in the configuration commands.
2. Clock Rates**:** I manually applied clock rate 64000 to bring the line protocols up.

7. Conclusion

This project successfully demonstrated the integration of static and dynamic routing protocols. By manipulating Administrative Distances, I was able to achieve a network that offers the redundancy of OSPF while strictly adhering to specific traffic engineering policies via Static Routing.